

ACADEMIC NETWORK INFRASTRUCTURE DESIGN

Open-Ended Lab Task & Practical Cisco Packet Tracer Implementation

Course: Computer Networks Tool: Cisco Packet Tracer Port Syntax: 0/0 & 0/1 Format

Architecture: Inter-VLAN Routing (802.1Q) [🔗](#) Status: Implemented & Verified

TASK 1: NETWORK REQUIREMENTS ANALYSIS

The institute requires an isolated, secure, and scalable local area network (LAN) for two key operational departments: **Administration** and **Academic**. The practical lab setup utilizes 4 representative host workstations connected through a core switch and inter-VLAN router.

Device Name	IP Address	Subnet Mask	Default Gateway	Department / VLAN
PC-Admin-1	192.168.10.2	255.255.255.0	192.168.10.1	Admin (VLAN 10)
PC-Admin-2	192.168.10.3	255.255.255.0	192.168.10.1	Admin (VLAN 10)
PC-Acad-1	192.168.20.2	255.255.255.0	192.168.20.1	Academic (VLAN 20)
PC-Acad-2	192.168.20.3	255.255.255.0	192.168.20.1	Academic (VLAN 20)

TASK 2: TOPOLOGY SELECTION & JUSTIFICATION

The **Hierarchical Star Topology** was selected for this design. End devices connect directly to central access switch ports, and the switch connects via a single trunk link to the core router.

Fault Isolation: A failure in any single workstation or cable does not affect other nodes on the network.

Ease of Expansion: New PCs can be added to VLAN 10 or VLAN 20 simply by connecting to open switch access ports without modifying the core topology.

Logical Segmentation: Keeps administrative data broadcasts isolated from student/academic networks.

TASK 3: NETWORK CABLING & PHYSICAL INFRASTRUCTURE PLAN

Link Connection	Cable Type	Port Designation	Description
PC to Switch Links	Copper Straight - Through (Cat6)	Fa0/1 - Fa0/20	Connects end workstations (PC-Admin, PC-Acad) to FastEthernet switch ports.
Switch to Router Link	Copper Straight - Through (Cat6)	Gig0/1 (SW) to Gig0/0 (RTR)	802.1Q Trunk Link carrying VLAN 10 & VLAN 20 tagged traffic across sub-interfaces.

TASK 4: SWITCH DESIGN & CONFIGURATION (VLANs & TRUNKING)

The switch SW-MAIN is configured with VLAN 10 for Administration and VLAN 20 for Academic departments. Gigabit Ethernet port Gig0/1 is configured as an 802.1Q trunk port connecting to router port Gig0/0.

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname SW-MAIN
Switch(config)#^
% Invalid input detected at '^' marker.

Switch(config)#hostname SW-MAIN
SW-MAIN(config)#vlan 10
SW-MAIN(config-vlan)#name ADMIN
SW-MAIN(config-vlan)#exit
SW-MAIN(config)#vlan 20
SW-MAIN(config-vlan)#name ACADEMIC
SW-MAIN(config-vlan)#exit
SW-MAIN(config)#interface range fastEthernet 0/1 - 10
SW-MAIN(config-if-range)#switchport mode access
SW-MAIN(config-if-range)#switchport access vlan 10
SW-MAIN(config-if-range)#no shutdown
SW-MAIN(config-if-range)#exit
SW-MAIN(config)#interface range fastEthernet 0/11-20
SW-MAIN(config-if-range)#switchport mode access
SW-MAIN(config-if-range)#switchport access vlan 20
SW-MAIN(config-if-range)#no shutdown
SW-MAIN(config-if-range)#exit
SW-MAIN(config)#interface gigabitEthernet 0/1
SW-MAIN(config-if)#switchport mode trunk
SW-MAIN(config-if)#switchport trunk allowed vlan 10,20
SW-MAIN(config-if)#no shutdown
SW-MAIN(config-if)#exit
SW-MAIN(config)#end
SW-MAIN#
%SYS-5-CONFIG_I: Configured from console by console
write memory
Building configuration...
[OK]
SW-MAIN#
```

```
SW-MAIN>show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gig0/2
10	ADMIN	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10
20	ACADEMIC	active	Fa0/11, Fa0/12, Fa0/13, Fa0/14 Fa0/15, Fa0/16, Fa0/17, Fa0/18 Fa0/19, Fa0/20
1002	fdi-default	active	
1003	token-ring-default	active	
1004	fdiinet-default	active	
1005	trnet-default	active	

```
SW-MAIN>
```

```
SW-MAIN>show interface trunk
```

Port	Mode	Encapsulation	Status	Native vlan
Gig0/1	on	802.1q	trunking	1
Port	Vlans allowed on trunk			
Gig0/1	10,20			
Port	Vlans allowed and active in management domain			
Gig0/1	10,20			
Port	Vlans in spanning tree forwarding state and not pruned			
Gig0/1	10,20			

```
SW-MAIN>
```

TASK 5: ROUTER CONFIGURATION & IP ADDRESSING SCHEME

Router-on-a-Stick inter-VLAN routing is enabled on physical port GigabitEthernet 0/0 of RTR-CORE using sub-interfaces Gig0/0.10 and Gig0/0.20.

Device / Interface	Sub-Interface	Encapsulation	IP Address	Subnet Mask	Purpose / Role
RTR-CORE (Gig0/0)	GigabitEthernet 0/0	Physical Link	Unassigned	N/A	Physical link to Switch Gig0/1
RTR-CORE (Gig0/0.10)	Gig0/0.10	dot1Q 10	192.168.10.1	255.255.255.0	Default Gateway for Admin Subnet
RTR-CORE (Gig0/0.20)	Gig0/0.20	dot1Q 20	192.168.20.1	255.255.255.0	Default Gateway for Academic Subnet

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname RTR-CORE
RTR-CORE(config)#interface gigabitEthernet 0/0
RTR-CORE(config-if)#no shutdown

RTR-CORE(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
exit
RTR-CORE(config)#interface gigabitEthernet 0/0/0.10
%Invalid interface type and number
RTR-CORE(config)#interface gigabit
% Incomplete command.
RTR-CORE(config)#interface gigabitEthernet 0/0.10
RTR-CORE(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.10, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to up
encapsulation dot1Q 10
RTR-CORE(config-subif)#ip address 192.168.10.1 255.255.255.0
RTR-CORE(config-subif)#description Default Gateway for VLAN 10 ADMIN
^
% Invalid input detected at '^' marker.

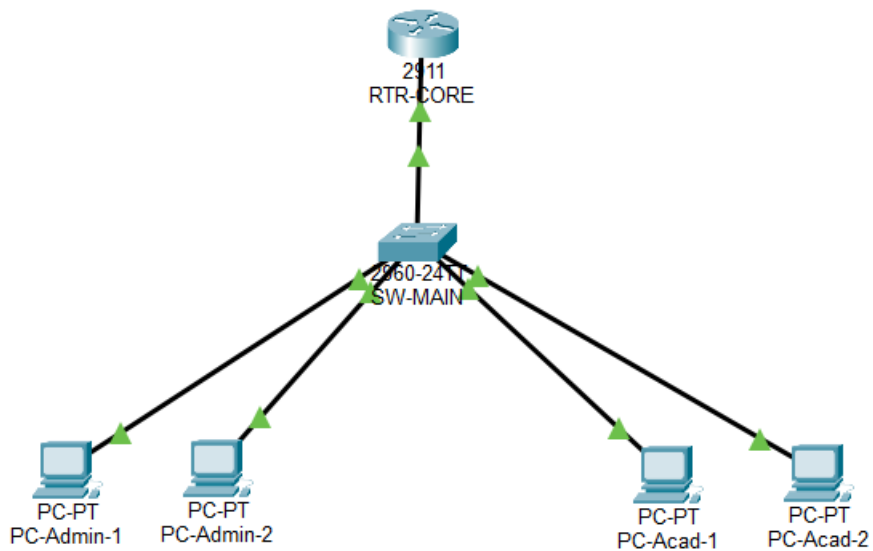
RTR-CORE(config-subif)#description Default Gateway for VLAN 10 ADMIN
RTR-CORE(config-subif)#exit
RTR-CORE(config)#interface gigabitEthernet 0/0.20
RTR-CORE(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.20, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to up
encapsulation dot1Q 20
RTR-CORE(config-subif)#ip address 192.168.20.1 255.255.255.0
RTR-CORE(config-subif)#description Default Gateway for VLAN 20 ACADEMIC
RTR-CORE(config-subif)#exit
RTR-CORE(config)#end
RTR-CORE#
%SYS-5-CONFIG_I: Configured from console by console
write memory
Building configuration...
[OK]
RTR-CORE#
```

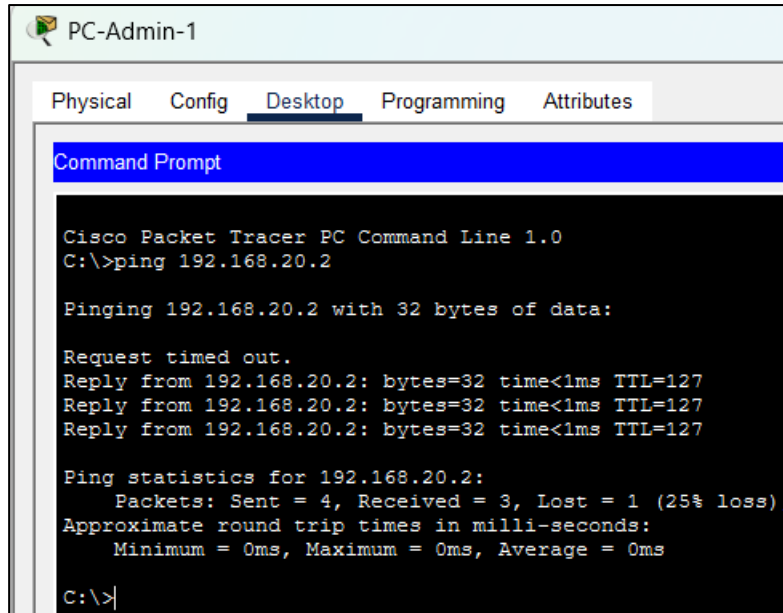
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Basic Topology



TASK 7: Inter-VLAN Ping Test Verification



TASK 8: FUTURE IMPROVEMENTS & SECURITY ENHANCEMENTS

Port Security Implementation: Configure static MAC address binding (``switchport port-security``) on switch access ports to prevent rogue laptops from connecting to the institute network.

Inter-VLAN Access Control Lists (ACLs): Implement extended ACLs on router interface ``Gig0/0.20`` to restrict student computers from accessing administrative server subnets while allowing general internet and gateway access.

Redundancy & Link Aggregation: Implement LACP EtherChannel between switches and deploy HSRP (Hot Standby Router Protocol) across dual routers for high availability.